

## High temperatures predict altered gut microbiome composition, metabolic capacity, and immune interactions in a wild primate species

--Manuscript Draft--

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| <b>Question</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Response</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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All the data and computer code used for this manuscript have been deposited in a GitHub repository that will be made public upon acceptance. Raw sequence data generated by 16s rRNA amplicon sequencing and shotgun metagenomics will be deposited on NCBI upon acceptance.

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December 11<sup>th</sup>, 2025

Dear Editor,

We are submitting our manuscript, “High temperatures predict altered gut microbiome composition, metabolic capacity, and immune interactions in a wild primate species,” to *PLOS Biology*. Here, we determine the influence of natural thermal variation on gut microbiome composition and metabolic capacity in a wild population of capuchin monkeys (*Cebus imitator*). We find that high temperatures are associated with altered community-level diversity, taxonomic composition, and metabolic capacity consistent with physiological responses to heat stress. We also show that the microbiome’s response to high temperatures is linked to host immunity and may affect gut-immune interactions. Previous experimental work has demonstrated that microbiomes play a key role in determining the thermal tolerances of their host species, but this research has largely been conducted in laboratory settings on endotherms and livestock, and offers limited insight into how species respond to natural thermal variation. Our results provide rare evidence from a natural population of primates that changes to the gut microbiome at high temperatures may contribute to physiological heat stress by compromising essential microbiome functions. Our results also highlight an important pathway by which rising temperatures due to climate change may threaten wild animal populations, particularly in tropical species with narrow thermal tolerances.

This manuscript describes original work and is not under consideration by any other journal. All authors approved the manuscript and this submission. All protocols involved in this study were previously approved by the University of Michigan IACUC (#PRO00011898) and comply with the relevant legal requirements governing animal research in the United States of America and Costa Rica.

Thank you for considering our manuscript. We appreciate your time and look forward to your response.

Sincerely,

A handwritten signature in black ink that reads 'Jordan Lucore'.

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# High temperatures predict altered gut microbiome composition, metabolic capacity, and immune interactions in a wild primate species

Running title: Heat alters primate gut microbiomes

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**Abstract:** Rising temperatures due to climate change are impacting the function of critical physiological processes in wild animals. Heat stress may compromise the gut microbiome's ability to support essential host functions, highlighting an important pathway by which climate change could negatively affect numerous species. However, the relationship between the microbiome and heat stress has been studied primarily in experimental contexts and remains poorly understood in ecologically relevant environments. Here, we examine how natural thermal variation influences gut microbiome taxonomic composition and functional capacity in a wild population of white-face capuchin monkeys. We find that high temperatures are associated with altered microbiome community-level diversity, taxonomic composition, and metabolic capacity consistent with physiological responses to heat stress. We also show that the microbiome's response to high temperatures is linked to host immunity and may affect microbiome-immune interactions. These results suggest that rising temperatures may compromise essential microbiome functions and highlight a potential mechanism by which climate change could negatively impact species through their microbiomes.

**Keywords:** gut microbiome, neopterin, heat stress, endotherm, climate change, wild primate

## Introduction:

Rising temperatures due to climate change are predicted to push species beyond their evolved thermal tolerances and challenge them to adapt to hotter climates (1, 2). Efforts to evaluate the physiological consequences of heat stress in wild animals have shown that high temperatures can impair aerobic capacity, reducing oxygen availability to essential systems and compromising

function (3, 4). Much of this research has focused on ectotherms, whose lack of physiological thermoregulation makes them particularly vulnerable to high temperatures (4–6). However, mounting evidence suggests that endotherms are also physiologically susceptible to heat (7, 8), and evolved thermoregulatory mechanisms may be increasingly insufficient to cope with prolonged periods of extreme heat. Recently, research has turned to assessing the role of host-associated microbiomes in shaping how species respond to heat stress. Animals have evolved complex relationships with the microbial communities (i.e., microbiomes) that live on and inside them to the extent that hosts rely on their microbiomes to support diverse aspects of their physiology. Accordingly, microbiomes may play a key role in determining the thermal tolerances of their host species (9–12) because heat stress can deplete microbiome communities, and is associated with negative phenotypic, survivorship, and fitness outcomes (13, 14). While this research highlights that heat is an important determinant of host-microbiome relationships, it has largely been conducted in laboratory settings and offers limited insight into how these dynamics function in ecologically relevant contexts.

The gastrointestinal tract is home to a particularly dense microbial community, the gut microbiome, that plays a crucial role in regulating host responses to environmental change (15), making it an ideal system to explore how species may respond to rising temperatures. High temperatures have been shown to alter community composition (16) and are linked to changes in host digestion, metabolism, and immune function (17–20), but these relationships have yet to be demonstrated in wild populations. A common consequence of heat stress is redox imbalance—the accumulation of reactive oxygen species (ROS) that can damage cells (21, 22), compromise the epithelial barrier (23), and trigger immune activation (24). Gut microbes typically help maintain redox balance (25–27), but heat-induced changes in community composition have been linked to reduced antioxidant metabolism (28, 29) and increased epithelial damage (30). These redox shifts may disrupt oxygen gradients, favoring facultative anaerobes over obligate anaerobes. This is supported across a variety of vertebrate species where high temperatures are consistently associated with declines in Firmicutes and Bacteroidota (20, 31, 32), phyla composed primarily of obligate anaerobes lacking robust ROS defenses (33, 34), and increases in Proteobacteria (16), a phylum dominated by facultative anaerobes that thrive in higher-oxygen conditions (35, 36).

The loss of obligate anaerobes during heat stress could affect the relationship between the gut microbiome and the immune system. Many obligate anaerobes ferment dietary fiber and produce short-chain fatty acids (SCFAs) (37–39), which support anti-inflammatory mechanisms that promote immune tolerance of the microbiome (40–43). A loss of these obligate anaerobes could result in reduced SCFAs and impact the immune system. Experimental studies evaluating the effects of acute heat stress in livestock show that exposure to high temperatures is associated with reduced SCFA concentrations (19), changes in the levels of circulating cytokines, and altered gut microbiome composition (17, 44), suggesting that heat-driven compositional shifts may disrupt microbiome-immune interactions and increase the risk of gut inflammation. These dynamics are likely to play an essential role in shaping animal responses to changing climates. However, interactions between heat stress and the microbiome have been studied almost exclusively in experimental contexts that examine the effect of acute periods of moderate heat stress [e.g., ~8–10 hours of exposure to temperatures ~32–34 °C, (45, 46)] and have focused either on ectotherms [e.g., amphibians (18, 47), fish (28), reptiles (32, 48)] or on domestic livestock [e.g., pigs (30), cows (17, 19, 44), poultry (29, 49)]. Therefore, we have little

understanding of how this system functions in wild populations in response to natural thermal variation.

To address this knowledge gap, here, we assess how temperature predicts gut microbiome composition and metabolic function in wild capuchin monkeys, *Cebus imitator*, in Costa Rica. We focus on evaluating the response to cumulative thermal load by averaging ambient temperature two weeks before sample collection, rather than attempting to measure short-term responses that are difficult to isolate in wild populations. Importantly, this predictor is also associated with thermal responses in immune performance in this population, supporting its relevance for studying the consequences of high temperatures on physiology (8). We also evaluate the interplay between temperature, the gut microbiome, and host immunity. Specifically, we measure host immunity using urinary neopterin, a byproduct of macrophage activity during the cellular immune response (50). Macrophages play a central role in host containment of the gut microbiome (51), and microbial metabolites, including SCFAs and saturated fatty acids, regulate macrophage activity to maintain host tolerance and prevent chronic inflammatory disorders (40, 52). Therefore, neopterin is a valuable tool for examining microbiome-immune homeostasis.

We analyzed 11 months of paired fecal and urine samples from 36 individually identified white-faced capuchin monkeys living in tropical dry forest fragments in northwestern Costa Rica at the Taboga Forest Reserve. Tropical dry forests are highly seasonal and defined by marked changes in ecology between the dry and wet seasons (53). During the dry season, lack of rainfall leads to food and water scarcity that has physiological consequences for capuchins (54) and can influence gut microbiome composition (55). Because the dry season is also characterized by high temperatures, this can make it difficult to disentangle the effects of heat from those of resource scarcity. However, our study population differs from most others in that we find no evidence of seasonal differences in physiological stress responses (56) or immunity (8), suggesting that these monkeys are not exposed to the same seasonal stressors. This is likely because our study site is highly disturbed and bordered by secondary forest and agriculture, where capuchins can exploit anthropogenic food and water sources even during the dry season (57). Consequently, studying this population provides a rare opportunity to isolate temperature-driven changes in the microbiome from those driven by broader changes in seasonal ecology. We characterized microbiome composition using 16S rRNA gene amplicon sequencing (n = 277), and characterized microbiome functional capacity using shotgun metagenomic sequencing (n = 71). Our results highlight the crucial role of temperature in shaping essential gut microbiome functions in a wild endotherm species and demonstrate how temperature is related to microbiome-immune interactions.

## Results:

### Temperature predicts gut microbiome taxonomic composition and functional capacity

#### *Community-level results*

We found a strong relationship between temperature and both taxonomic and functional community-level diversity in the gut microbiome. To assess this relationship, we compared a baseline model that accounted for known predictors of microbiome composition (sex, social group, diet, and season) to a model that included both the baseline predictors and temperature (Fig. 1S). We measured alpha diversity using Shannon diversity index and observed richness (the total number of Amplicon Sequence Variants, ASVs, or pathways in a sample), and characterized beta diversity using the first three principal components (PC1–PC3) from a

principal component analysis (PCA) to capture major axes of variation in community composition. For taxonomic community composition, the model including temperature provided the best fit (controlling for model complexity) across all community-level indices, accounting for 100% of the model weight for every principal component and alpha diversity metric (Fig. 1B). The strong relationship between temperature and principal components 1–3 suggests that temperature drives shifts in taxonomic composition that are substantial enough to affect community-level beta diversity. Both alpha diversity indices showed a reliable negative relationship with temperature (Shannon  $\beta_{\text{temp}} = -0.49$ , Observed  $\beta_{\text{temp}} = -0.49$ )(Fig. 1A). Even though the effect sizes were small compared to the principal components, alpha diversity reliably declined with increasing temperatures (Fig. 2S). This suggests that higher temperatures are associated with reductions in both taxonomic richness and evenness.

For functional capacity, the temperature model provided the best fit for every principal component and the total number of observed metabolic pathways, accounting for at least 96% of the model weight. In contrast, the baseline model provided the best fit for Shannon diversity, accounting for 77% of the model weight (Fig. 1D). Therefore, temperature was associated with shifts in individual metabolic pathways substantial enough to affect community-level beta diversity, but the relationship between temperature and functional alpha diversity was less clear. Temperature was negatively associated with the total number of unique pathways (Observed:  $\beta_{\text{temp}} = -35.05$ ), but the magnitude and direction of this effect were uncertain due to the wide confidence intervals that include zero. The effect size for Shannon diversity ( $\beta_{\text{temp}} = -0.27$ ) was much smaller, and the confidence intervals also included zero (Fig. 1C). Together, these results suggest that higher temperatures may reduce the number of unique metabolic pathways but are not associated with changes in pathway evenness.

#### *Individual taxon and metabolic pathway results*

Temperature was also related to the abundance of individual taxa at both the family and genus levels. The temperature model provided the best fit (controlling for complexity) for 93 out of 163 taxa, accounting for at least 50% of the model weight. The majority of taxa, 82 out of 93, declined in abundance with rising temperatures (Table 1S). Ancestral state reconstruction performed using the temperature model weights suggests that this is a conserved microbial trait for *C. imitator*, particularly in key phyla such as Actinobacteria, Bacteroidota, and Firmicutes. In contrast, phyla such as Campylobacterota and Proteobacteria appear less sensitive to temperature over evolutionary time (Fig 2).

Temperature was also related to the abundance of microbial metabolic pathways. The temperature model provided the best fit for 169 out of 383 total pathways, accounting for at least 50% of the model weight (Table 2S). To facilitate interpretation, we grouped 103 of these pathways into five functional categories based on shared biological roles and their apparent relationship with temperature: lipid metabolism, oxygen-driven metabolism and fermentation, redox maintenance, nucleotide mobilization, and immune regulation (Fig 3). Patterns in these categories suggest a metabolic restructuring that may help microbes respond to rising temperatures. For example, we found an increase in fatty acid synthesis pathways like oleate, cis-vaccenate, palmitoleate, stearate, petroselinate, and gondoate, and lipid remodeling pathways such as the mevalonate pathway and downstream isoprenoid lipid synthesis (Fig 3A), that may help maintain cytoplasmic membrane integrity at high temperatures (58, 59). However, we also found a decrease in pathways involved in lipid catabolism and recycling, including fatty acid  $\beta$ -oxidation, fatty acid salvage, and phospholipid turnover, which likely impairs membrane repair

mechanisms (59, 60) and may compromise integrity during extended periods of high temperatures. Further, we observed an increase in anaerobic pathways, including the reductive TCA cycle and several involved in glycolysis, alongside a decrease in pathways that support aerobic metabolism, including the pyruvate dehydrogenase complex, the TCA cycle, and glyoxylate cycle (Fig 3B). This may reflect a response to changing oxygen gradients with rising temperatures to maintain energy production (61). The anaerobic pathways primarily metabolized simple carbohydrates and coincided with a decline in complex polysaccharide fermentation, likely reflecting taxonomic shifts with rising temperatures (37).

We also observed declines in pathways that support redox maintenance and cycling (Fig 3C), including the synthesis of cofactors such as B vitamins (62, 63), coenzymes (64), antioxidant precursors to glutathione (65), NADPH production, quinone synthesis, siderophore-mediated iron transport (66, 67), and detoxification pathways for aromatic compounds and aldehydes (68). In conjunction, we observed an increase in nucleotide and translation-associated metabolism pathways (Fig 3D), such as purine and pyrimidine de novo synthesis, nucleotide salvage and recycling (69, 70), queuosine synthesis, and S-adenosylmethionine salvage (71). This coordinated increase in nucleotide and translation-associated metabolism may reflect the consequences of redox imbalance at high temperatures because the accumulation of ROS can lead to DNA damage (72). Some pathways that support redox maintenance increased with rising temperatures (Fig 3D), including the synthesis of antioxidant precursors and vitamin B12 (73), coenzyme metabolism (74), and degradation of vitamin-like compounds (75), which may reflect a compensatory effort to restore redox balance.

Rising temperatures were also associated with a decline in pathways that produce metabolites known to support anti-inflammatory signaling (Fig 3E), including synthesis of the SCFAs butyrate and propanoate (42, 76), catabolism of monounsaturated fatty acids like oleate (77), and polyamine production from arginine degradation and ornithine synthesis (78). However, these shifts were accompanied by an increase in select pathways that may help compensate for the loss of microbial immune regulation, including the synthesis of the saturated fatty acids (5Z)-dodecenoate and L-tryptophan biosynthesis, which leads to the production of indole derivatives that suppress pro-inflammatory cytokine signaling (79).

#### Neopterin is related to the gut microbiome's response to temperature

##### *Community-level results*

To assess if neopterin concentration is associated with the gut microbiome's relationship with temperature, we compared the temperature model that included known predictors of microbiome composition plus temperature to a model that included the baseline predictors and an interaction effect between temperature and neopterin concentration (Fig 1S). For taxonomic composition, the model including the interaction effect between temperature and neopterin concentration provided the best fit (controlling for complexity) for every principal component, accounting for at least 60% of the model weight (Fig 3S). Principal component 1 was the only component with confidence intervals that did not include zero. This result, along with the high model weight for every principal component, suggests that the relationship between temperature and beta diversity is associated with neopterin. In contrast, the temperature model provided the best fit for both alpha diversity metrics, accounting for at least 86% of the model weight. These results, combined with the small effect sizes (Shannon  $\beta_{\text{temp}*\text{neopterin}} = -0.10$ , Observed  $\beta_{\text{temp}*\text{neopterin}} = -0.13$ ) and confidence intervals that include zero (Fig 3S), suggest that the

relationship between taxonomic alpha diversity and temperature is independent of neopterin concentration.

For metabolic functional composition, the interaction model provided the best fit for every principal component and observed pathways, accounting for at least 98% of the model weight, while the temperature model provided the best fit for Shannon diversity, accounting for 63% of the model weight (Fig 3S). Even though principal component 3 was the only component where the confidence intervals did not include zero, the large effect sizes for all three components (PC1  $\beta_{\text{temp}*\text{neopterin}} = 17.60$ , PC2  $\beta_{\text{temp}*\text{neopterin}} = -7.66$ , PC3  $\beta_{\text{temp}*\text{neopterin}} = -11.44$ ) plus the model weights suggest that the interaction between neopterin and temperature has a meaningful influence on metabolic functional composition. The effect size and model weight for observed pathways (Observed  $\beta_{\text{temp}*\text{neopterin}} = 72.78$ ) suggest that neopterin may be associated with the relationship between the total number of distinct metabolic pathways and temperature. However, it is unlikely that neopterin is associated with the relationship between temperature and pathway evenness due to the small effect size (Shannon  $\beta_{\text{temp}*\text{neopterin}} = 0.46$ ), confidence intervals that include zero, and model weight for Shannon diversity.

#### *Individual taxon and metabolic pathway results*

Neopterin concentration was also associated with the relationship between temperature and the abundance of individual taxa. The interaction model provided the best fit for 20 of the 93 temperature-associated taxa, accounting for at least 50% of the model weight in each case (Table 3S). To interpret the interaction effect, we plotted the relationship between temperature and abundance across three discrete ranges of neopterin values (Fig 4). These results indicated two contrasting effects: (1) the relationship between temperature and abundance became more negative (i.e., the slopes decrease) at higher neopterin concentrations (Fig 4A), and (2) the relationship between temperature and abundance became more positive (i.e., the slopes increase) at higher neopterin concentrations (Fig 4B & C). The majority of taxa (12 out of 20) exhibited decreasing slopes at higher neopterin concentrations. These taxa were almost exclusively from the phyla Bacteroidota (5 taxa) and Firmicutes (6 taxa), and included two known SCFA producers, *Parabacteroides* and *Alistipes* (80, 81). The taxa that exhibited increasing slopes (8 out of 20) showed two distinct patterns: one where a positive slope became steeper when neopterin was high, and another where a negative slope changed to positive when neopterin was high. These taxa included two SCFA producers, *Phascolarctobacterium* and *Eubacterium* (82, 83), and the family *Helicobacteraceae*, which commonly includes facultative anaerobes (84).

Neopterin concentration was also associated with the relationship between temperature and the abundance of individual metabolic pathways. The interaction model provided a better fit for 65 out of 116 pathways tested, accounting for at least 50% of the model weight in each case (Table 4S). To interpret the interaction effect, we plotted the relationship between abundance and temperature across two discrete ranges of neopterin values. These results indicated the same two effects as the taxonomic analysis (1) where slopes decreased at higher neopterin (Fig 5A-C) and (2) where slopes increased at higher neopterin (Fig 5D & E). The pathways that exhibited decreasing slopes showed three distinct patterns: one where a positive slope changed to negative, another where a positive slope became less positive, and one where a negative slope became steeper. This group included pathways involved in anti-inflammatory signaling, including synthesis of monounsaturated fatty acids (77) and tryptophan, as well as those typically associated with obligate anaerobes, including complex carb fermentation (Flint et al., 2012) and Gram-positive cell wall biosynthesis characteristic of obligate anaerobes from the Firmicutes

phylum (85). The majority of pathways (37 out of 65) exhibited increasing slopes at higher neopterin concentrations. There were two patterns of increasing slopes: one where a negative slope changed to positive when neopterin was high, and another where a positive slope became steeper when neopterin was high. These groups included pathways commonly associated with facultative anaerobes, including the glyoxylate cycle (86), Rubisco shunt (87), ubiquinone biosynthesis (88), enterobacterial common antigen biosynthesis (89), and Lipid A biosynthesis (90). They also included pathways involved in redox maintenance, including the synthesis of B vitamins (62, 63), menaquinol (64), and NADPH (67).

#### Discussion:

Our study had two main goals: (1) to assess how temperature influences gut microbiome composition and function in wild, white-faced capuchins, and (2) to evaluate whether temperature-driven changes to the gut microbiome are related to host immunity. We found strong evidence that temperature is associated with community-level diversity, taxonomic composition, and metabolic capacity that may indicate the capuchins are undergoing periods of heat stress. Moreover, the microbiome's taxonomic and functional responses to temperature are linked to differences in urinary neopterin concentrations, suggesting that immunity is related to the gut microbiome's response to heat.

#### Gut microbiome response to temperature

Our data suggest that the gut microbiome's response to high temperatures may affect its role in conferring colonization resistance—the ability to protect the host from exogenous pathogens. Colonization resistance arises from direct competition of commensals with pathogens for resources and microbial support for host immunity (91). Community-level analysis reveals that taxonomic alpha diversity declines with increasing temperatures (Fig 2S), suggesting that heat may impair the gut microbiome's ability to resist colonization by reducing the diversity of taxa that are available to compete with pathogens (92). High temperatures are also associated with the decline in the Bacteroidota and Firmicutes phyla, which contain taxa known to support host immunity such as *Clostridiaceae*, *Lactococcus*, and *Barnesiella* (Fig 2)(93), which may leave hosts more vulnerable to infection. However, functional alpha diversity appears more resilient to increasing temperatures (Fig 1), likely due to functional redundancy—when taxonomically distinct microbes perform similar functional roles (94). This redundancy may buffer the loss of taxonomic diversity and help preserve colonization resistance at high temperatures because remaining taxa fill enough functional niches to outcompete pathogens. Although we did not observe an increase in potentially pathogenic taxa, the phyla Proteobacteria and Campylobacteria, which are known to contain gut pathogens (36, 95), are less sensitive to heat compared to other phyla. Therefore, sustained high temperatures may continue to erode both taxonomic and functional diversity, eventually increasing the host's vulnerability to infection.

Metabolic restructuring in response to rising temperatures suggests that the gut microbiome undergoes functional shifts that may support community stability. For example, the increase in lipid synthesis and remodeling pathways (Fig 3A) could reflect cytoplasmic membrane restructuring, a process bacteria use to preserve membrane stability at high temperatures (58, 96). However, these changes may be insufficient to maintain membrane integrity during extended periods of high temperatures because the observed decline in lipid catabolism and recycling pathways likely compromises membrane repair mechanisms (59). Limited repair capacity could leave gut microbes more vulnerable to peroxidative damage, a

common consequence of heat stress driven by an accumulation of ROS that damages the lipid components of cell membranes (21, 97). Similarly, the observed increase in anaerobic metabolic pathways and the decline in aerobic pathways (Fig 3B) likely reflect a response to maintain energy production under changing oxygen gradients due to redirected blood flow during heat exposure (61). However, these anaerobic pathways are primarily involved in simple carbohydrate metabolism typical of facultative taxa (98, 99), and coincide with a decline in complex polysaccharide fermentation typical of obligate taxa (37–39). Therefore, this shift is likely driven by the relative persistence of facultative versus obligate anaerobes at high temperatures. Since fiber fermentation creates SCFAs, which are crucial energy sources (100) and immunomodulators (41, 42), extended periods of high temperatures may eventually cause a decline in overall energy availability and affect immune function. While these metabolic shifts suggest that capuchin gut microbes actively respond to heat in ways that may support community stability, the decline in membrane repair and inferred loss of obligate fermenters indicate that this response is unlikely to be sustainable over longer periods of high temperatures.

The taxonomic response to temperature suggests that changes in gut physiology at high temperatures affect the persistence of obligate anaerobes compared to facultative anaerobes. There are broad declines in two phyla known to include obligate anaerobes, Bacteroidota and Firmicutes, including essential obligate taxa such as *Bacteroides*, *Parabacteroides*, *Clostridium*, *Ruminococcus*, *Eubacterium*, *Blautia*, *Anaerostipes*, and *Alistipes* (Fig 2). In contrast, Proteobacteria, comprised of primarily facultative taxa (35, 36), are relatively stable in response to rising temperatures. These patterns mirror compositional changes in response to heat stress in a wide range of vertebrates [e.g., fish (101, 102), reptiles (32, 48), amphibians, birds (20), and mammals (103)]. The decrease in obligate anaerobes is likely due to redox imbalance induced by high temperatures, where the production of reactive oxygen species (ROS) exceeds the antioxidant capacity of the gut environment. Obligate anaerobes possess limited enzymatic defenses against ROS, and are therefore less able to tolerate changes in the oxygen environment compared to facultative taxa (33, 34). Although direct markers of redox balance are needed to assess the gut's redox environment, the observed decline in redox-maintenance pathways (Fig 3C) suggests that rising temperatures impair microbial antioxidant capacity, which can disrupt the redox environment (25–27). We also observe an increase in DNA repair pathways (Fig 3D) that suggests cellular damage from the accumulation of ROS (72). Together, these results provide compelling evidence that capuchins experience periods of redox imbalance at high temperatures that may affect the relative abundance of facultative versus obligate anaerobes.

Further, a decline in obligate anaerobes during heat stress may disrupt the gut microbiome's ability to regulate host immunity by impairing key immunomodulatory functions. Obligate anaerobes are essential SCFA producers that help regulate intestinal immunity and maintain host tolerance towards gut microbiomes, including the promotion of anti-inflammatory regulatory T-cell activity (41, 76) and the suppression of proinflammatory cytokine signaling (42, 77). The decrease in SCFA-producing pathways, particularly butyrate and pyruvate (Fig 3E), likely reflects the loss of obligate anaerobes at high temperatures due to redox imbalance and may decrease host tolerance towards gut microbiomes. The resulting increase in immune activity may exacerbate redox imbalance because activated immune cells, particularly phagocytes, release ROS as part of the inflammatory response (104).

These results may be driven by confounding variables related to seasonal ecology, such as changes in diet, ranging patterns, social structure, activity budgets, or water sources that can influence microbiome composition and function (55, 105, 106). However, many of our results

mirror those from experimental studies evaluating the effects of high temperature on the microbiome, indicating that we have likely captured a true temperature effect rather than an indirect signature of seasonal ecology. This is likely because we assessed cumulative exposure to high temperatures rather than short-term fluctuations that are more likely to covary with other ecological variables. We also focused on interpreting broad patterns across taxonomic groups and metabolic pathways (Figs. 2 & 3) rather than interpreting individual features that may be more sensitive to ecological confounds.

Importantly, our results resemble those associated with heat stress in experimental work, including a decrease in obligate anaerobes relative to facultative anaerobes (16), evidence of redox imbalance (28, 29), and changes in immunomodulatory function (17, 19, 44). Together, these findings suggest that the capuchins may be experiencing heat stress at the upper limit of their thermal range that could compromise essential microbiome functions. These effects also likely reinforce one another during extended periods of high temperatures via reciprocal interactions linking redox balance, microbial composition, and immune function (107, 108)(Fig 3F).

#### Gut microbiome response to temperature is related to microbiome-immune interactions

Declines in the abundance of obligate anaerobes with temperature may impact host immunity. For example, stronger declines in obligate anaerobes at high temperatures, such as *Parabacteroides* and *Alistipes*, are associated with higher neopterin levels (Fig 4A). These taxa contribute to anti-inflammatory signaling through the production of SCFAs (80, 81), suggesting that their loss may affect immune tolerance towards gut microbes. Similarly, stronger declines in metabolic pathways that produce anti-inflammatory immunomodulators are also associated with higher neopterin. This includes the synthesis of monounsaturated fatty acids (77) and tryptophan, a compound primarily synthesized by obligate anaerobes (109, 110) (Fig 5A & B). Conversely, high neopterin could also contribute to a stronger decline in obligate anaerobes at high temperatures, likely because increased macrophage activity exacerbates heat-driven redox imbalance. Taxa that exhibit stronger declines when neopterin is high are primarily from the Bacteroidota and Firmicutes phyla, which include many obligate anaerobes (Fig 4A). We also observed a stronger decline in pathways commonly associated with obligate anaerobes including complex carb fermentation (37) and Gram-positive cell wall biosynthesis characteristic of obligates from the Firmicutes phylum (85)(Fig 5A).

In contrast, facultative anaerobes may be more resilient to these conditions. For example, *Helicobacteraceae* exhibits a stronger increase in abundance when neopterin is high (Fig 4B), suggesting they are better able to tolerate redox imbalance, and may exploit niches vacated by the loss of obligates. Similarly, several pathways commonly associated with facultative anaerobes exhibit a stronger increase at high neopterin levels and may help these taxa actively respond to changes in the gut environment (Fig 5D). For example, the glyoxylate cycle and Rubisco shunt both increase under high neopterin, which likely supports energy conservation under oxygen and energy-limited conditions (86, 87). In addition, redox maintenance pathways such as ubiquinone biosynthesis also increase which helps facultative anaerobes neutralize ROS (88). Synthesis of enterobacterial common antigen (ECA) also increases, which is characteristic of *Enterobacteriaceae* (89). Conversely, an increase in facultative anaerobes at high temperatures could also affect host immunity. For example, lipid A synthesis, which is characteristic of many facultative taxa, increases at higher neopterin and forms pathogen-associated molecular patterns (PAMPs) that are recognized by Toll-like receptors (TLRs) and

can initiate pro-inflammatory signaling (90)(Fig 5D). *Helicobacteraceae* is also associated with inflammatory diseases and may promote macrophage activation by damaging the gut epithelium and disrupting the mucosal barrier (111).

Therefore, the decline of obligate anaerobes relative to facultative taxa at high temperatures may drive, and be driven by, increased immune activity in the gut. Although we cannot conclude directionality, these findings suggest that gut-immune interactions are an integral component of the gut microbiome's response to heat and may increase susceptibility to gut inflammation during prolonged high temperatures. These results may also reflect physiological heat stress in the capuchins, given the observed relationship between taxonomic shifts and immune activity that mirrors experimental findings, and the potential inflammatory consequences of long-term heat exposure (16, 17, 19, 44). Future work should incorporate additional immune biomarkers that directly assess inflammation, as neopterin primarily reflects macrophage activation and represents only one of the many mechanisms linking immunity and the gut microbiome. Continued research should also assess whether these effects have measurable impacts on host health and fitness, providing critical insight into how climate warming may influence wildlife through gut-immune pathways.

#### Conclusion:

Our results highlight the pivotal role of temperature in shaping the gut microbiome and its relationship with host immunity in a wild primate species. Many of the observed taxonomic and metabolic shifts resemble those characteristic of experimentally induced heat stress, including declines in obligate anaerobes relative to facultative anaerobes, evidence of redox imbalance, and altered immunomodulatory functions, suggesting that the capuchins may be experiencing heat stress that could compromise essential microbiome functions. We also found that gut-immune interactions are linked to the gut microbiome's response to heat and may increase susceptibility to gut inflammation during prolonged high temperatures. These results highlight the crucial role of temperature in shaping essential gut microbiome functions in a wild endotherm species and offer insight into how rising temperatures due to climate change may threaten species resilience through their microbiomes.

#### Methods:

##### Site information and data collection

We collected data from a wild population of *C. imitator* living in the Taboga Forest reserve in Guanacaste, Northwestern Costa Rica. The Taboga Forest Reserve is a 516-ha tropical dry forest fragment and forms an important piece of a fragmented biological corridor connecting the Tempisque River Basin to the Guanacaste Mountains (57). The Capuchinos de Taboga Research project has collected daily behavioral and endocrine data on the three groups of white-faced capuchins since 2018. All individuals are individually identified and habituated to human presence. Individual age is calculated by birth date for individuals younger than 7 years of age; for individuals older than 7 years, age was estimated based on body condition by a field team member (J.C.O.) with more than two decades of experience observing wild white-faced capuchins. A team of two or more trained field assistants collected data between 5:00 AM and 6:30 PM including detailed observations of feeding and foraging behavior using scan sampling. Scans occurred every 30 minutes for 10 minutes, where feeding and foraging observations of individually identified monkeys were recorded. Individuals were recorded multiple times in one scan if they were observed engaging with different food items. Plant food items were identified on the family or genus level.

Between September 2022 and July 2023, we opportunistically collected fecal and urine samples in conjunction with behavioral data. All samples were collected from identified individuals who appeared healthy upon visual inspection. We placed fecal samples into 4 mL of 98% ethanol immediately after defecation (112). To avoid contamination, we wore gloves and only collected feces that had not contacted the soil. We discarded feces contaminated with urine to prevent contamination from the urogenital microbiome (113). We collected urine samples using a clean catch method (54) to avoid rainfall and fecal contamination. We also collected urine from leaves or the ground if it appeared free of soil or fecal contamination. We immediately stored fecal and urine samples in a portable cooler and transferred them to a –20°F freezer at the end of each field day. We kept fecal samples under refrigeration until shipment to the Amato Microbiome Lab at Northwestern University in Evanston IL for DNA extraction and sequencing. Urine samples remained in the freezer until we conducted assay analysis onsite in the Capuchinos de Taboga field laboratory (56).

Temperature observations for this analysis came from the Palo Verde meteorological station, located at the Organization for Tropical Studies (OTS) Palo Verde research station (10.345°N, 85.339°E), approximately 20.81 km southwest of the Taboga Forest Reserve. Like Taboga, Palo Verde is one of Central America's last remaining tropical dry forest fragments (114). Palo Verde temperature data have been used in previous studies of the Taboga white-faced capuchin population (116)(8), given the habitat similarity and proximity to Taboga. The station records maximum and minimum air temperature every 15 minutes.

#### Neopterin analysis

We measured the neopterin concentration of the urine samples using a commercially available competitive exclusion neopterin enzyme-linked immunosorbent assay (Neopterin ELISA, Ref. RE59321, IBL International GMBH, Hamburg, Germany) in the Capuchinos de Taboga field laboratory. The assay has been validated for use in the Taboga capuchin population (115). We used specific gravity to control for sample water volume and diluted samples using manufacturer assay buffer. We ran the assays using manufacturer protocol, see (8, 115) for details.

#### Sample processing for 16S rRNA amplicon and shotgun metagenomic sequencing DNA extraction

We conducted all lab work at the Amato Microbiome Lab at Northwestern University, Evanston IL. We extracted DNA from fecal samples using the DNeasy PowerLyzer PowerSoil Kit (Qiagen, Germantown, MD, USA), with minor modifications to manufacturer protocol. Specifically, we incubated samples at 65°C for 10 minutes after adding the C1 and PowerBead solutions, refrigerated the C2 and C3 incubation steps, and prewarmed the C6 solution before the final elution. These modifications align with other protocols designed for fecal samples containing high levels of plant fiber (116–118).

#### 16s rRNA amplicon sequencing and data processing

We used 16s rRNA amplicon sequencing to describe the gut microbiome taxonomic composition. We amplified the hypervariable V4-V5 region of the 16s rRNA gene using the 515F/926R primers from the Earth Microbiome Project ([www.earthmicrobiome.org](http://www.earthmicrobiome.org)) and a dual-indexing approach (119); see previously described protocol for details (120). We used negative controls to monitor contamination during the DNA extraction and PCR amplification. We

purified and normalized PCR products using SequelPrep Normalization and sequenced them on the Illumina MiSeq V4 platform at the Rush Genomics and Microbiome Core Facility.

We analyzed the resulting data using the Quantitative Insights into Microbial Ecology (QIIME2) platform (121), v2023.9. Pair-end sequences from 900 samples yielded 28,869,202 raw sequence reads (average 32,077 per sample, range of 7 to 64851 reads). We trimmed forward and reverse reads to 216 bases and 206 bases, respectively, to remove low-quality base pairs using a minimum Phred score of 33. We quality-filtered and denoised sequences using the Divisive Amplicon Denoising Algorithm 2 (DADA2)(122) via the q2-dada2 pipeline, which infers Amplicon Sequence Variants (ASVs) by identifying exact sequences. We assigned taxonomy to ASV's using the q2-feature classifier (123) against version 2022.10 of the Greengenes2 database (124, 125) by applying the classify-sklearn model. We removed sequences assigned to chloroplasts and mitochondria using the q2-taxa-filter-table to retain only microbial sequences for downstream analysis. All negative controls contained fewer than 507 reads per sample (average of 29), and we excluded them after visually confirming that their taxonomic composition did not indicate contamination. We also excluded one fecal sample that contained fewer than 1,000 reads to avoid bias from low-coverage data. After filtering, we retained 797 samples, which contained an average of 16,898 reads (range: 4,944–38,941).

We imported the ASV count and taxonomy files generated by QIIME2 into R using the {qiime2R} package (126) and saved them as a *phyloseq* object using the {phyloseq} package (127). Most ASV's were found at low frequencies; out of 11,916 total ASV's, 89% were only found in one sample, and 16% were not assigned to a phylum. To avoid analyzing taxa that are artifacts of the sequencing process (128), we removed ASV's found in less than 0.5% of total samples. After filtering, we removed 102 additional ASV's that were not assigned at the phylum level, resulting in 1159 total remaining ASV's. Most of the remaining ASV's could be assigned to class (100%), order (95%) and family (93%) levels but assignment decreased substantially at the genus (60%) level.

### *Shotgun metagenomic sequencing and data processing*

We used metagenomic sequencing to characterize the functional capacity of the gut microbiome in a subset of 95 samples with sufficient DNA concentration for library preparation. We prepared sequencing libraries using the NEBNext Ultra™ II FS DNA PCR-free Library Prep Kit for Illumina (New England Biolabs, Ipswich, MA, USA). We followed the manufacturer protocol for low inputs (50ng to 99ng) and selected library inserts around 350bp. We pooled the libraries and sequenced them on the Illumina NovaSeq platform at the Rush Genomics and Microbiome Core Facility. Pair-end sequences from 95 samples yielded 2,902,714,548 raw sequence reads (average 15,277,445 per sample, range of 531,829 to 25,348,795 reads).

We quality-filtered the raw metagenomic sequences using KneadData (v0.12.0) which uses Trimmomatic to remove adaptors, low-quality reads, and overrepresented or repetitive sequences, and Bowtie2 to remove host-derived sequences (129). We ran KneadData using the standard parameters SLIDINGWINDOW:4:20 and MINLEN:90 to trim reads when the average quality score dropped below 20 and remove reads shorter than 90bp. To remove host-associated reads we used the GRCh38.p13 human reference genome (RefSeq accession: GCF\_009914755.1) which has been used to filter host genome data in previous capuchin metagenome studies (55). KneadData yielded 931,326,495 cleaned reads (average 98,034,376 per sample, range of 178,711 to 22,386,906 reads). We concatenated the forward and reverse reads and used HUMAnN3 (v3.9) for functional profiling. Gene families were assigned using the

Uniref50 database and regrouped into MetaCyc reactions. The resulting table reported the abundance of each unique metabolic pathway normalized by the number of gene families contributing to that pathway. We imported the pathway abundance table into R and removed the unmapped values, which represent sequences that could not be assigned to any known gene family, as well as unintegrated values, which correspond to gene families not associated with any known pathway. After filtering, 548 unique metabolic pathways remained. We performed all subsequent analyses using R v4.3.1.

#### Metadata assembly

We matched each 16s rRNA and shotgun metagenomic sample with the relevant metadata to facilitate subsequent analyses. We paired samples with the average neopterin value from a 7-day window, encompassing three days before, the day of, and three days after sample collection. Our time window provides a general measure of immune performance associated with the microbiome composition, enabling us to assess the relationship between the two. We also averaged the maximum temperature (collected every 15 minutes) 15 days before sample collection. This time frame allowed us to evaluate the relationship between the microbiome and long-term trends in temperature. This temperature predictor has also been associated with changes in immune performance in this population (8).

We also paired each sample with all diet observations from the same individual within two weeks before sample collection, following time frames used in previous white-faced capuchin gut microbiome studies (55, 106). We quantified diet diversity by modeling diet richness (i.e., the total number of unique food items recorded) as a function of the total number of feeding observations. We then used the residuals as a measure of diet diversity, with negative values indicating lower-than-expected diversity and positive values indicating higher-than-expected diversity (130). We calculated the percent consumption of each food type by dividing the total number of feeding observations for each food category (animal, flower, fruit, fungi, insect, leaves, nectar, nurse, pith, seeds) by the total number of feeding observations. We then plotted each food type against temperature to identify diet categories that vary with temperature to control for these factors when assessing the direct effect of temperature on the microbiome. We found that percent fruit consumption has a strong, positive relationship with temperature (Fig 4S) and controlled for this in subsequent analyses. After matching each sample with the appropriate metadata, we were left with 277 16s rRNA samples and 71 shotgun metagenomic samples.

#### Community-level composition analysis

To evaluate how gut microbiome taxonomic composition and functional capacity are related to temperature, we generated metrics of community composition including alpha and beta diversity. To measure alpha diversity, we calculated two widely used indices: Shannon diversity index (which accounts for both richness and evenness) and observed richness (the total number of ASVs or pathways in a sample) using the `estimate_richness()` function from the `{phyloseq}` package. To measure beta diversity (i.e., community dissimilarity between samples), we generated Aitchison distances by computing the Euclidean distance between pairwise samples after performing a centered log-ratio (clr) transformation on the raw count data (131). We added a pseudo count to all zeros before the clr transformation using the `cmultRepl()` function from the `{zCompositions}` package, which uses Bayesian-multiplicative replacement to preserve the structure of the data and minimize distortion compared to adding a fixed value (132). We used

Aitchison distances because microbiome datasets are compositional, meaning that observed read counts are constrained by sequencing depth and do not reflect absolute DNA abundances (133). Unlike traditional beta diversity metrics that were designed for absolute counts (e.g., Bray-Curtis, Unifrac, Jensen-Shannon divergence), the clr transformation projects samples into Euclidean space where distances reflect log-ratio relationships among taxa, avoiding misinterpretations that may arise from using absolute differences to interpret compositional data (134). We then performed a Principal Components Analysis (PCA) on the Aitchison distance matrix and extracted the scores for the first three principal components (PC1–PC3) to capture major axes of variation in community composition. Components one through three were selected because they explained greater than 5% of the variation (Fig 5S).

To evaluate the relationship between temperature and community composition, we built mixed models using alpha diversity indices and the first three principal components as the outcome variables. We modeled each community metric twice, and compared model fit using Akaike information criterion adjusted for small sample sizes (AICc) (135, 136). The baseline model, *m\_main*, included predictors for social group, age, sex, rainfall, diet diversity and fruit consumption. The temperature model, *m\_full*, included all of the same predictors with an additional predictor for temperature (Fig 1S). Both models included a random intercept for individual to control for repeated sampling and an offset for the log-transformed number of reads to control for differences in sequencing depth between samples. We included fruit consumption as a predictor because it increased with rising temperatures (Fig 4S), and we wanted to control for this effect to better isolate the relationship between temperature and community features. We scaled all continuous predictors (i.e., age, rainfall, diet diversity, fruit consumption, temperature) for ease of coefficient interpretation. We fit all models as linear mixed models (LMMs) with a Gaussian distribution except for observed taxonomic richness, which was fit using a Gamma distribution with a log link function because the variable was continuous and positive with a long right tail. We used the model weights from the AICc comparison to evaluate the importance of temperature as a predictor of community composition.

Since rarefaction to account for differences in sequencing depth (i.e., choosing a fixed sequencing depth and randomly subsampling all samples to that depth) is strongly discouraged because it discards valuable data (137), we used alternative methods to leverage the full dataset (i.e., clr transformation, modeled offset for sequencing depth). Further, we chose to use an Information-theoretic approach for our analysis because, unlike null-hypothesis significance testing, it allows for the comparison of multiple alternative hypotheses to assess biological significance (138, 139). This allowed us to evaluate if adding temperature to our model improved the explanation of community-level taxonomic and metabolic diversity compared to other known predictors of microbiome composition.

#### Temperature differential abundance analysis

We also used a comparative modeling approach to identify taxa and metabolic pathways associated with temperature. For the taxonomic analysis, we aggregated the number of reads per taxa on the family and genus levels using the `tax_glom()` function in the `{phyloseq}` package. We used this to calculate the relative abundance of each taxon by dividing the number of reads per taxon by the total number of reads per sample and removed taxa with less than 0.2% average relative abundance across all samples to avoid analyzing extremely low-abundance taxa. This resulted in 134 genera and 79 families to be used for differential abundance testing.

We built two generalized linear mixed models (GLMMs) using a negative binomial distribution with a log link function and used individual family and genera abundances as our outcome variables. We used negative binomial GLMMs because they can effectively model over-dispersed count data with an abundance of zeros, a pattern typical of microbiome datasets. We modeled each taxon twice and compared model fit using AICc. The baseline model, *m\_main*, included predictors for social group, age, sex, rainfall, and fruit consumption. The temperature model, *m\_full*, included all of the same predictors with an additional predictor for temperature (Fig 1S). Both models included a random intercept for individuals to control for repeated sampling and an offset for the log-transformed number of reads to control for differences in sequencing depth between samples. Similar to our community feature analysis, fruit consumption was included to better isolate the relationship between temperature and individual taxa while accounting for diet components also related to temperature. However, we did not include diet diversity as a predictor to reduce model complexity and facilitate model convergence. We scaled all continuous predictors. Model weights were used to evaluate the importance of temperature as a predictor of individual taxon abundance. Taxa where *m\_full* held greater than 50% of the model weight were considered temperature sensitive. We then used the *m\_full* model weights of all genera included in the analysis to perform ancestral state reconstruction using the *ace()* function in the {ape} package. Internal node values represent the maximum likelihood estimates of the temperature model weight under a Brownian motion model of trait evolution.

The modeling approach for the metabolic pathway data mirrored that used for the taxonomic data, except we modeled clr-transformed abundances instead of raw counts. This allowed us to use a Gaussian distribution and omit the offset term for log-transformed read depth, as the clr transformation already accounts for differences in sequencing depth. We used clr-transformed abundances because pathway abundance values are normalized by the number of contributing gene families, meaning they are no longer raw counts and are therefore inappropriate for modeling with a negative binomial distribution. We modeled each of the 547 pathways twice, once with the baseline model, *m\_main*, and once with the temperature model, *m\_full* (Fig 1S). We compared the models using AICc. As in the taxonomic analysis, pathways for which *m\_full* accounted for more than 50% of the model weight were considered temperature sensitive.

#### Immune and temperature interaction analysis

To assess whether the relationship between temperature and community composition is associated with immune performance, we employed a second comparative modeling approach and used alpha diversity and the first three principal components as outcome variables. Similar to our previous approach, we modeled each community feature twice, once with each model, and compared model fit using AICc. The baseline model, *m\_immune\_main*, included the same predictors as *m\_full* (i.e., social group, age, sex, rainfall, fruit consumption, temperature, random effect for ID, log total reads offset). The interaction model, *m\_immune\_full*, included all of the same predictors with an additional interaction effect between temperature and neopterin (Fig 1S). The interaction effect allowed us to test if the relationship between community composition changes depending on neopterin levels. We scaled all continuous predictors, including neopterin. All models were fit as linear mixed models (LMMs) with a Gaussian distribution except for observed taxonomic richness, which was fit using a Gamma distribution with a log link function because the variable was continuous, positive, and left-skewed with a long right tail.

To assess whether the relationship between temperature and individual taxa or pathways is associated with immune performance, we employed the same model comparison approach but used taxon and pathways abundances as outcome variables. First, we filtered for taxa and pathways that were temperature-sensitive (i.e., *m\_full* held greater than 50% of the model weight). Then, we modeled each feature twice, once with *m\_immune\_main* and once with *m\_immune\_full*, and compared model fit using AICc. Importantly we fit the models with taxa as outcome variables as GLMMs with a negative binomial distribution, log link function, and included a log total reads offset. We fit the models with pathways as outcome variables as LMMs with a Gaussian distribution and did not include a log total reads offset. Features where *m\_immune\_full* held greater than 50% of the model weight were considered to have a relationship between temperature and abundance that changes depending on neopterin levels. We visualized these features using a coplot, where the relationship between feature abundance and temperature was plotted over distinct ranges of neopterin values.

Data availability statement: All the data and computer code used for this manuscript have been deposited in a GitHub repository that will be made public upon acceptance. Raw sequence data generated by 16s rRNA amplicon sequencing and shotgun metagenomics will be deposited on NCBI upon acceptance.

Author contributions statement:

Conception and design of the manuscript: J.M.L. and A.J.M.

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Field site funding, logistics, and management: M.E.B., J.C.B., T.J.B., and J.C.O.

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## Figures

**Figure 1:** Model comparison results assessing whether adding temperature to a base model of known microbiome predictors improves the explanation of community-level taxonomic and metabolic diversity. Panels **A** and **C** show the coefficients of the temperature predictor for community diversity metrics, including beta diversity (measured by principal components 1-3) and alpha diversity (measured by the Shannon Index and the observed number of ASVs or metabolic pathways). The thick bars represent 50% confidence intervals (CI), while the thin bars represent 95% CI. Panels **B** and **D** compare the model weights of the base model (in green) to the temperature model (in yellow). The weights of the base model plus the temperature model equal 100%. The top panels **A** and **B** show the results for community-level taxonomic diversity, while the bottom panels **C** and **D** show the results for metabolic pathway diversity. Results show a strong association between temperature and beta diversity for both taxonomic composition and metabolic capacity. In contrast, the association between temperature and alpha diversity differs by data type: taxonomic alpha diversity declines with rising temperature, whereas metabolic alpha diversity indicates a possible reduction in the number of unique pathways at higher temperatures but no relationship with pathway evenness.

**Figure 2:** Cladogram showing ancestral state reconstruction of temperature sensitivity for 97 genera with  $\geq 0.2\%$  average relative abundance. Branch color indicates model weight for the temperature model (includes base predictors plus temperature), with yellow representing high model weight (greater temperature sensitivity) and green representing low model weight. Genera where the temperature model accounts for  $\geq 50\%$  of the model weight are classified as temperature sensitive and are depicted with red (negative association) or blue (positive association) circles at the branch tips. Genera with  $< 50\%$  model weight are considered not temperature sensitive and are shown with grey circles. Most temperature-sensitive taxa decline in abundance with rising temperatures, especially within the Firmicutes and Bacteroidota phyla. In contrast, Proteobacteria and Campylobacterota appear less temperature sensitive compared to the other phyla.

**Figure 3:** Model results for the microbial metabolic pathways associated with temperature. Pathways are grouped into five functional categories (Panels **A-E**) based on shared biological roles and their associations with temperature. The thick bars represent 50% confidence intervals (CI), while the thin bars represent 95% CI. Summary annotations to the right of each panel describe the broad functional roles of the pathways within each category. Panel **F** visualizes the effects consistent with experimentally induced heat stress and how they may reinforce one another during extended periods of high temperatures.

**Figure 4:** The relationship between taxa abundance and temperature plotted across three distinct ranges of neopterin concentration. There are two main effects. Row **A** includes taxa for which the relationship between temperature and abundance becomes increasingly negative at higher neopterin concentrations. Rows **B** and **C** include taxa for which the relationship becomes increasingly positive, either through an increasingly positive slope (Row **B**) or a slope that switches from negative to positive (Row **C**). Taxa highlighted in the results and discussion are labeled on the right side of each row. A complete list of all taxa included in the plots is available in Supplementary Table 3.

**Figure 5:** The relationship between metabolic pathway abundance and temperature plotted across two distinct ranges of neopterin concentration. Pathway abundances are shown as centered log-ratio (clr)-transformed values. There are two main effects. Rows **A-C** include pathways for which the relationship becomes increasingly negative at higher neopterin, either through a slope that switches from positive to negative (Row **A**), a slope that becomes less positive (Row **B**), or an increasingly negative slope (Row **C**). Rows **D** and **E** include pathways for which the relationship between temperature and abundance becomes increasingly positive at higher neopterin, either through a slope that switches from negative to positive (Row **D**) or an increasingly positive slope (Row **E**). Pathways highlighted in the results and discussion are labeled on the right side of each row. A complete list of all pathways included in the plots is available in Supplementary Table 4.

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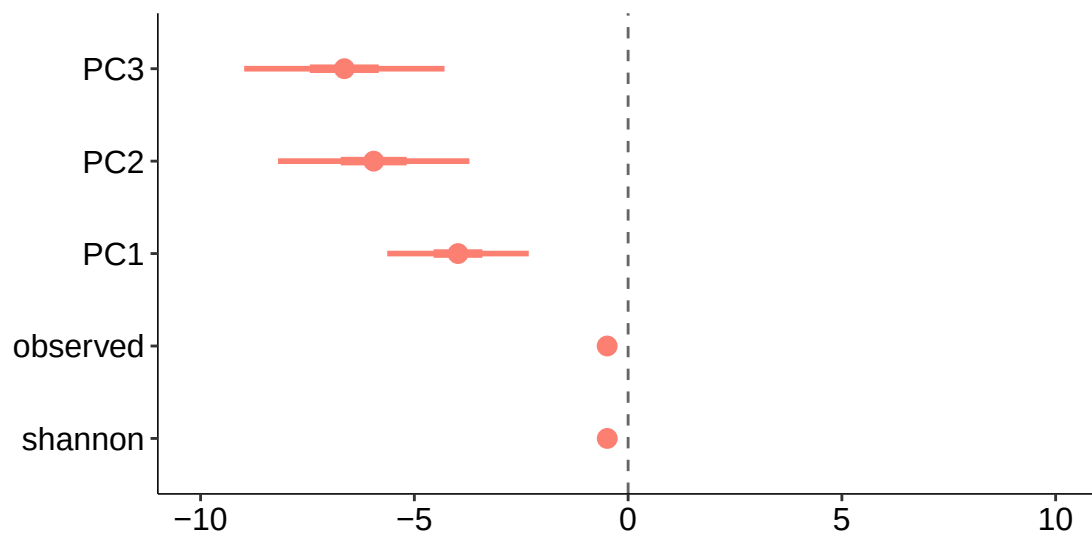
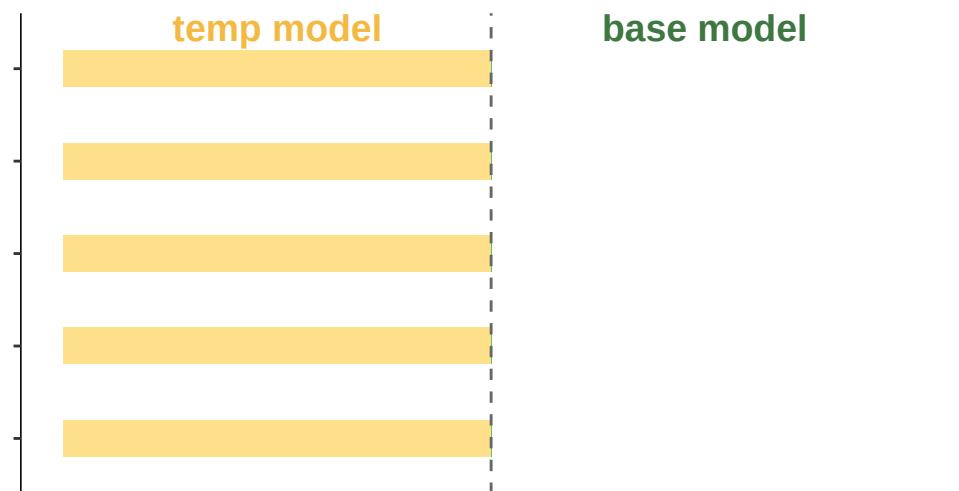
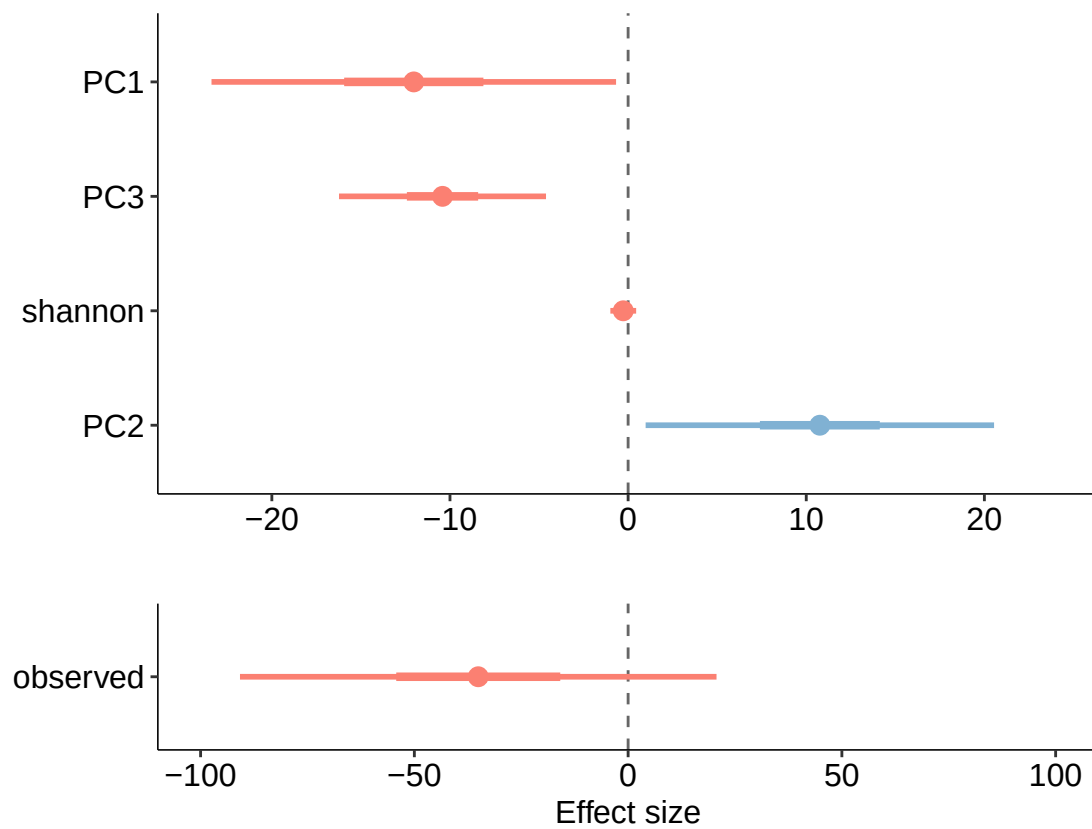
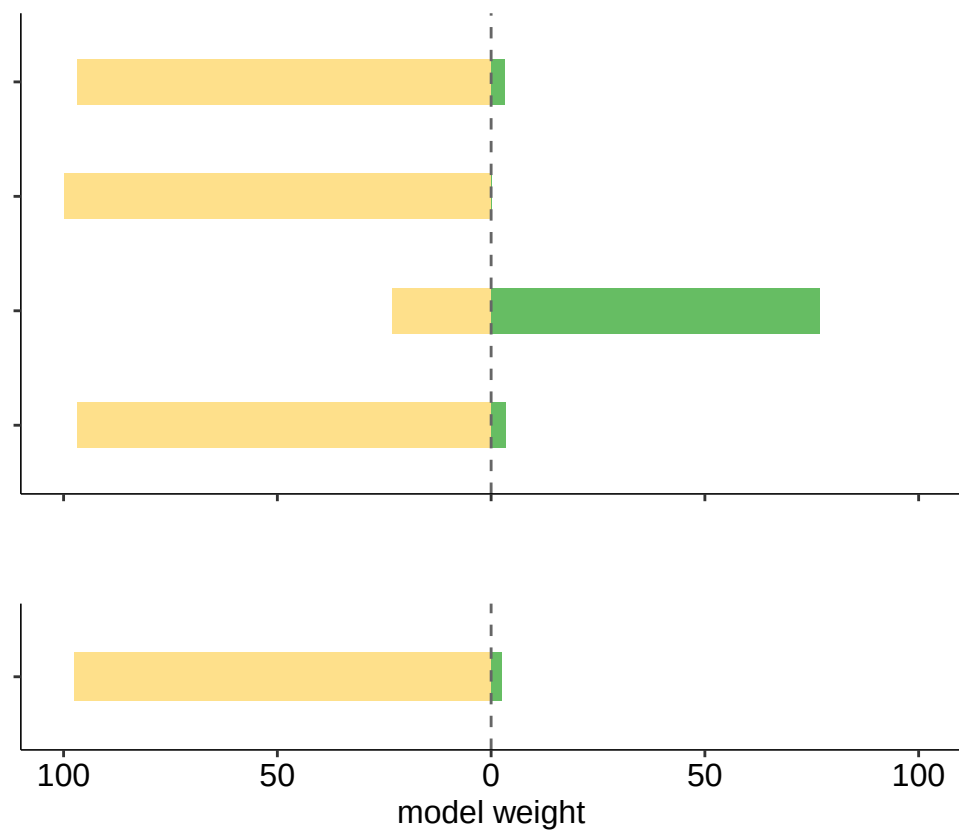
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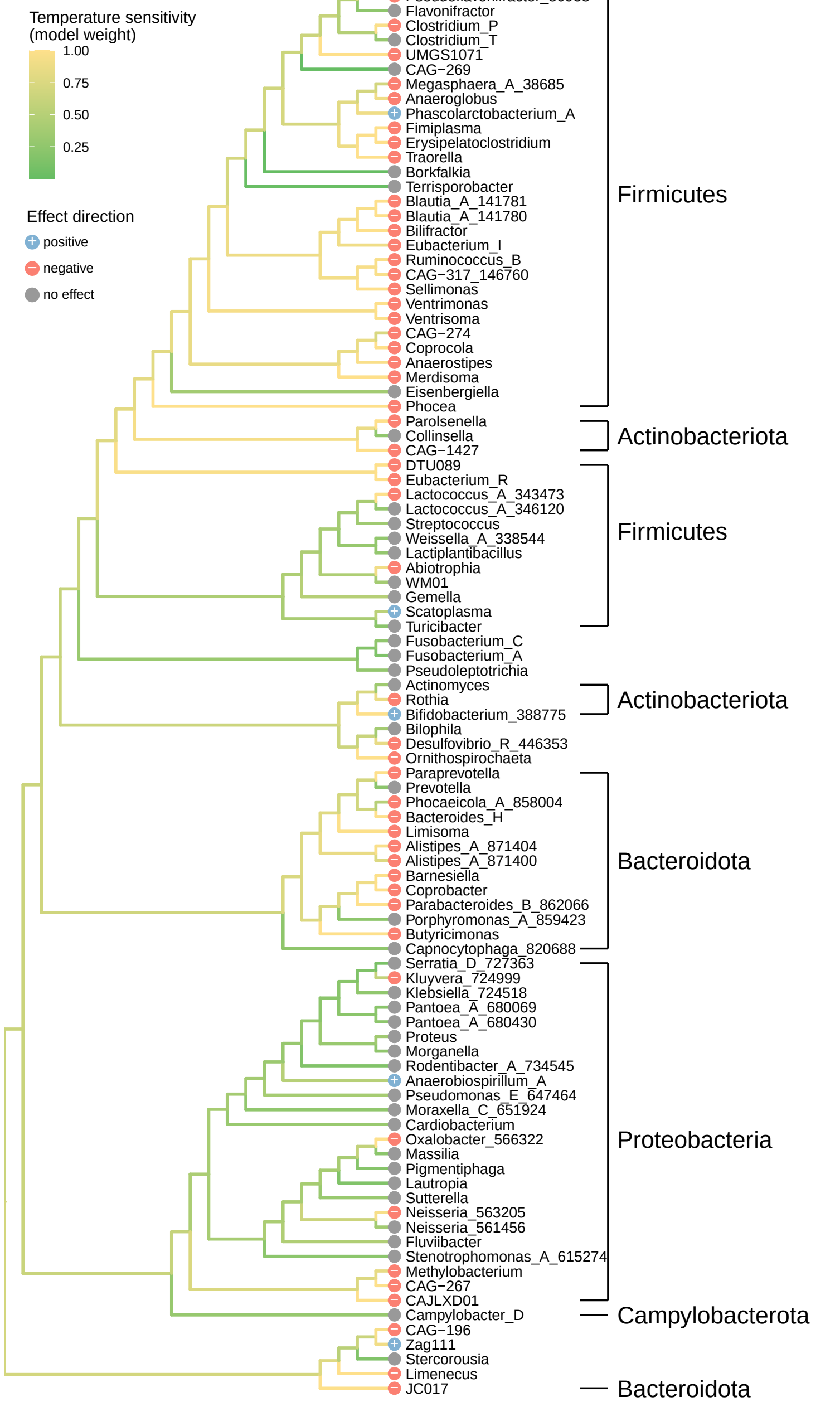
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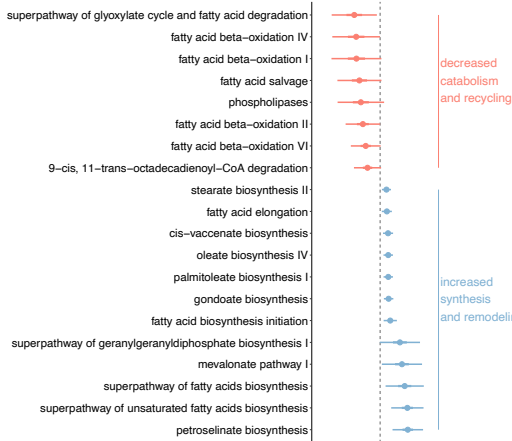
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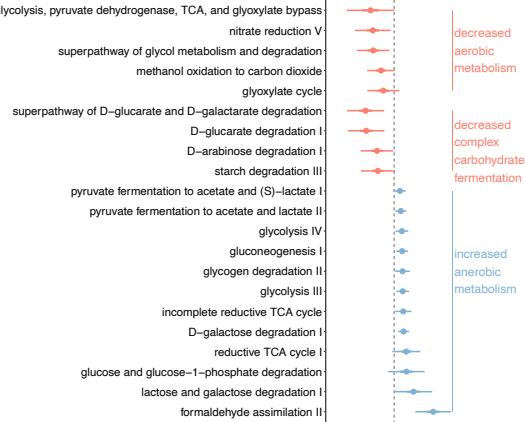
**A. Taxonomy temperature coefficients****B. Taxonomy model weights****C. Metabolic pathway temperature coefficients****D. Metabolic pathway model weights**



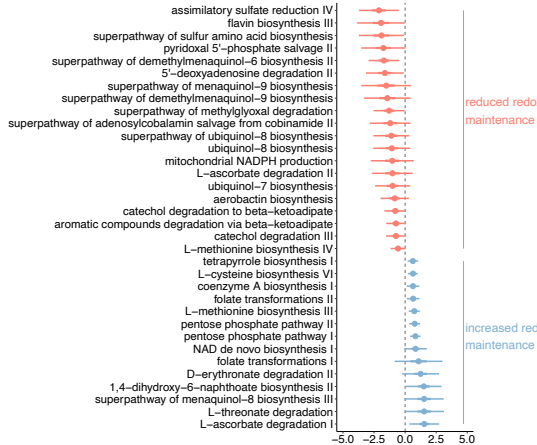
### A. Lipid metabolism



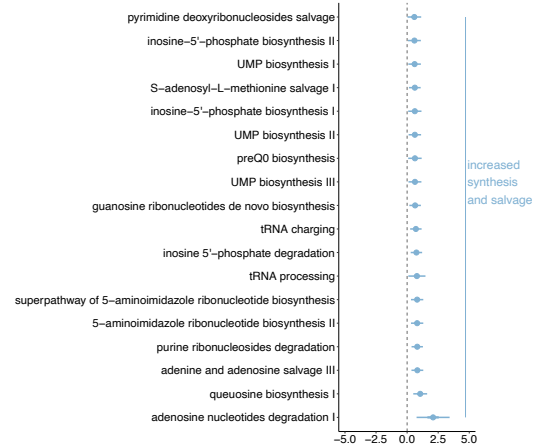
### B. Oxygen-driven metabolism and fermentation



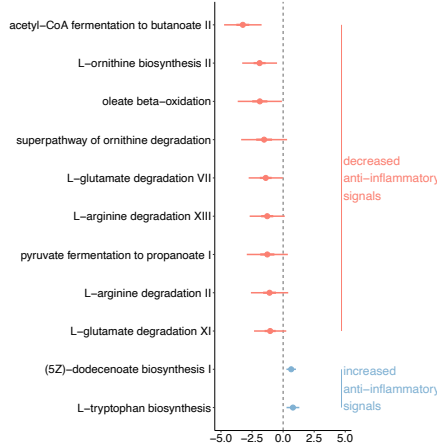
### C. Redox maintenance



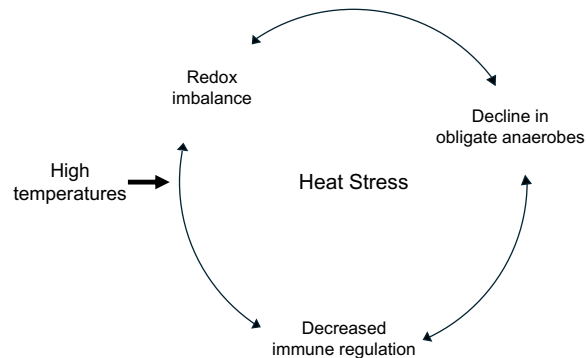
### D. Nucleotide mobilization



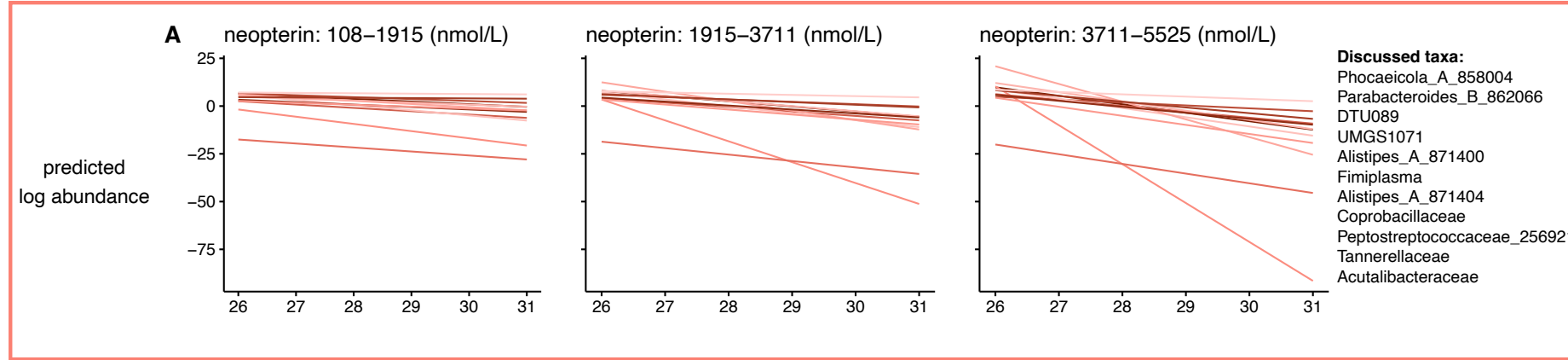
### E. Immune regulation



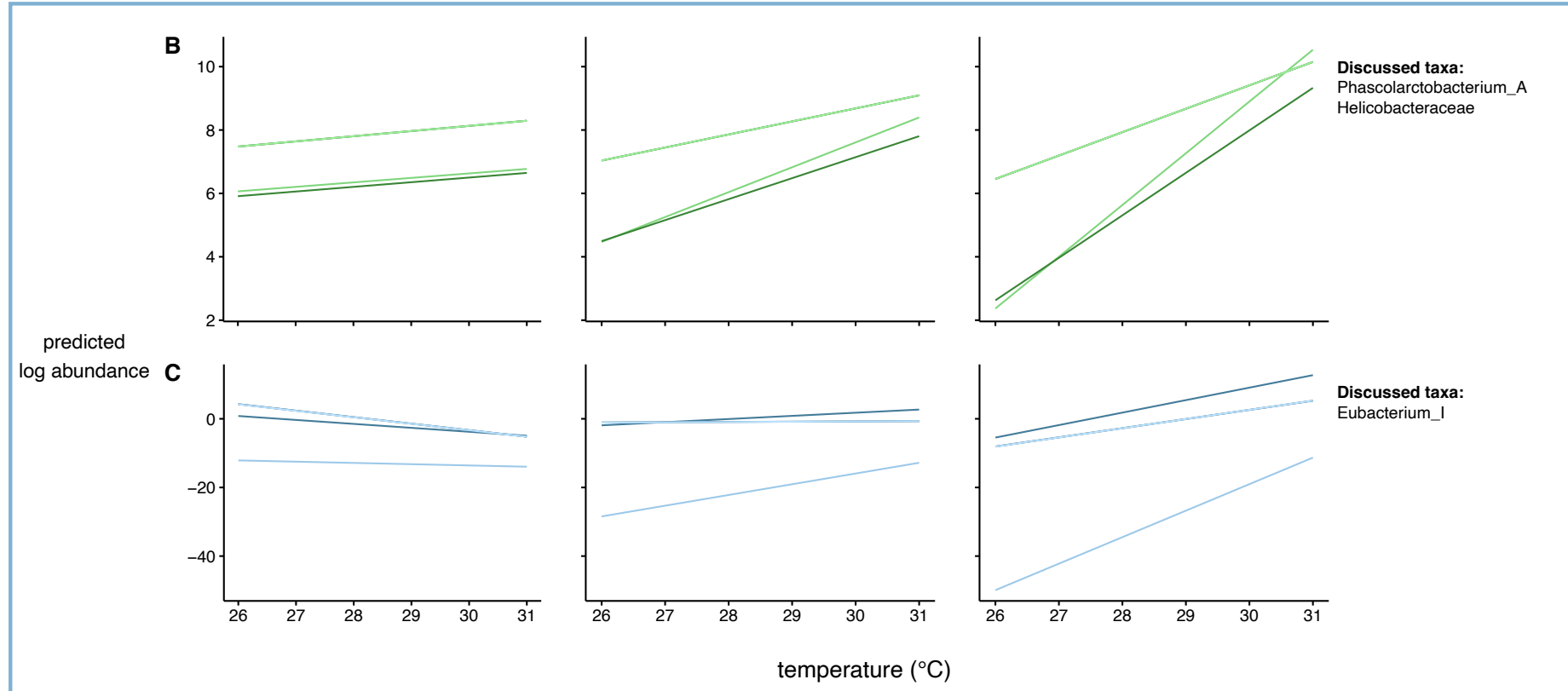
### F. Heat stress mechanism



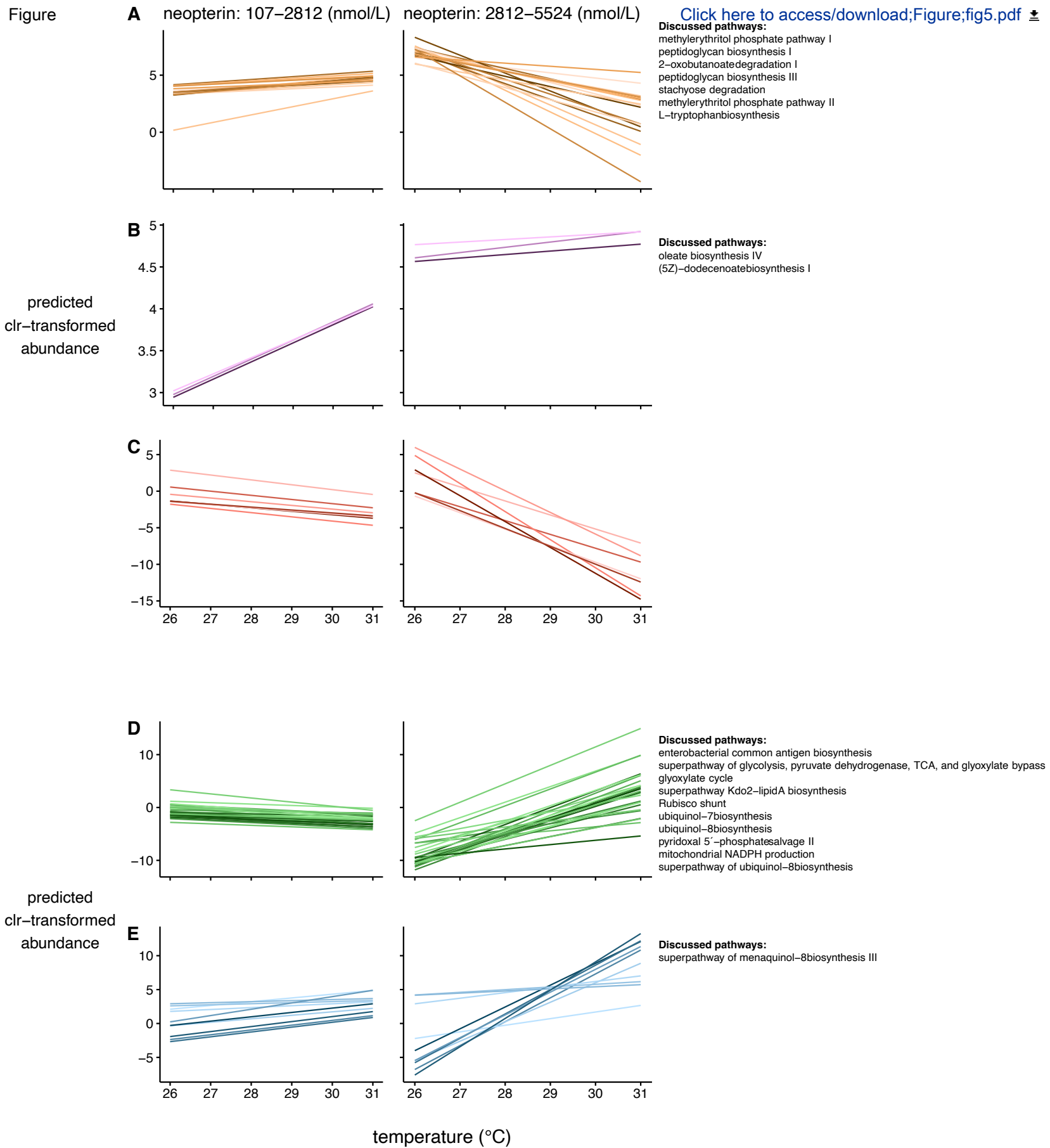
# Decreasing slopes



# Increasing slopes



Figure







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